

Testing for Carbon Leakage:
The Kyoto Protocol, Consumption-Based Emissions and the
Environmental Kuznets Curve, 1990 to 2023

Bachelor's Thesis

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Abstract

In terms of territorial carbon emissions, developed countries have been able to achieve reductions since 1990. It is, however, uncertain to what extent these drops are due to genuine decarbonisation processes or to the relocation of carbon-intensive production to countries with no climate restrictions. This research examines the second possibility using the case of Kyoto Protocol as a natural experiment. The Protocol imposed obligations to cut emissions on Annex B countries starting from the year 2005 while letting others do not adhere to any limits, thus being the situation under which carbon leakage is expected to occur. The researchers combined territorial and consumption-based fossil emissions data from Global Carbon Budget with that from World Bank and United Nations to create a 119-country panel (34 committed countries and 85 uncommitted ones) for the period 1990 to 2023. To assess the effect of commitment, the authors used two-way fixed-effects difference-in-differences model as well as event studies and joint tests of the pretreatment trends; also, robustness checks were conducted by removing transitional economies, by moving the treatment point to 2008, checking for the anticipation issues at the time of the accord signing, and by dropping large and non-large countries. There were three outcomes fixed in priority order before estimation. As for the first – considering the difference between emissions based on consumption and production as a part of production, the authors assessed the effect to be -0.070 (with the standard error equal to 0.073). This means that the estimated result is the opposite from what the hypothesis suggests and it is not statistically significant; the same holds true for secondary and robustness outcomes. Thus, the result can be defined as a very imprecise null; upon conducting tests of the pre-treatment trends for both main outcomes, the joint test rejected the hypothesis of parallel trends. Hence, there is no evidence that Kyoto has led to carbon leakage that could be disclosed in national accounts.

Keywords: carbon leakage; Kyoto Protocol; consumption emissions; difference-in-differences; Environmental Kuznets Curve

1 Introduction

1.1 Background

Between 1990 and 2023, CO₂ emissions in high income countries stopped increasing, in fact in some it started to decrease. At the same time, emissions rose steeply in middle-income countries. Post the final wave of globalization in 1990's and the creation of World trade Organization in 1995, trade also expanded. More of the goods bought by rich countries were manufactured somewhere else. Putting these facts side by side and a problem appears. If emissions are counted where goods are produced, a fall in a rich country's emissions only shows decarbonization if the carbon has not simply moved to the countries that now make its goods. This has significance for climate policy in that the commitments made at the international level under the United Nations Framework Convention on Climate Change are assessed based on territorial or production accounts. To achieve these commitments a country could either decrease emissions within its borders or simply increase imports of emission-based products that it produced before. However, only the first approach can reduce emissions. Consumption accounting allocates emissions to the point of consumption of goods, thus making the two approaches differ (Davis et. al, 2010) [10]. The difference between production-based and consumption-based emissions

of a country represents the "trade" or volume of CO₂ emitted by the country because of its trade activities. This is what this paper will attempt to analyze.

1.2 The puzzle: Decarbonization or Relocation?

According to the Environmental Kuznets Curve (EKC) theory, pollution increases as income increases during early stages of economic growth but decreases once income surpasses some level [15]. If we look at it strictly, the declining pollution in wealthier nations may imply that economic growth is self-correcting in terms of its environmental impact. However, there are two main objections to this theory. The first one pertains to econometrics, stating that the residuals in the model may change according to the specification and sample values. The second criticism is structural. It's possible that emissions reduction in developed countries comes not from the structural changes in production in those countries themselves, but from the transfer of polluting industries from developed countries to developing countries (Copeland and Taylor, 2004) [9]. The second criticism of EKC is supported by the research done by Peters et al. (2011), who shows that the amount of emissions embodied in goods being traded from developing countries to developed countries has substantially increased between 1990 and 2008. Both issues present themselves in this exploratory analysis of the thesis. Fixed effects estimate accounting for country effects and year effects reveal that emissions follow an inverted U-shaped curve with a turning point equal to \$38,700 in terms of production-based income and \$41,900 in terms of consumption-based income. Both of which are well within the range of observed incomes. Pooled quadratic fitting of the same data suggests that the turning point is much higher than those figures with such a large bootstrap confidence interval that it tells us nothing. Thus, how to estimate the turning point becomes an important issue. The more serious issue, though, is that a production-based inverted U-shaped curve shows not only genuine decarbonization but also relocation. One estimate of consumption-based emissions produces a similar turning point which, although indicative, does not come from an isolated approach. From that fitting, it is impossible to find the effect of climate policy on emissions, so another and more direct estimate needs to be undertaken.

1.3 The Kyoto Protocol as a natural experiment

The Kyoto Protocol was signed in 1997 and began to take effect in February 2005. It imposed limits on greenhouse gas emissions to its Annex B countries. It differed from Malaysia and Singapore, India, and China since countries listed as parties were not subjected to any binding agreement regarding emission reductions. Carbon leakage could occur in such cases. For this to happen binding target would make production in Annex B countries costly thus forcing businesses to emulate their means of production in jurisdictions where emissions don't experience any price impacts. Aichele and Felbermayr (2012, 2015) [1] used this asymmetry to postulate that Kyoto-induced emissions from jurisdictions that were obligated to follow the protocol were entering their territories in the form of imports. Nevertheless, studies conducted in various jurisdictions show contradictory results. For instance, so-called emissions trading scheme (ETS) in the European Union is found to have very low carbon leakage. Furthermore, estimates of carbon leakage rate differ drastically between model-based and empirical research. Thus, empirical evidence whether binding climate commitments were able to trigger a measurable carbon leakage phenomenon remains inconclusive.

1.4 Research question and hypotheses

The research question is: Did binding commitment under the Kyoto Protocol cause Annex B countries to become larger net importers of embodied carbon, relative to countries without binding targets? • The null hypothesis (H_0) is that Kyoto commitment had no effect on the consumption-production gap of Annex B countries relative to non-Annex B countries. • The alternative (H_1) is that commitment widened that gap, which is how carbon leakage would appear in national accounts. Three outcome measures were assigned a specific priority before any estimation took place. The primary outcome is the gap share, which is given by the difference between consumption-based emissions and production-based emissions divided by production-based emissions. The second outcome is the net transfer (in tonnes of CO₂ per person). The scaled inverse hyperbolic sine of net transfer is used for robustness as the transfer function has heavy tails. The order of priority allows the thesis to report all outcomes despite their signs. It also prohibits the selection of outcomes after the results are obtained. The design tests the possibility of leakage and does not start from the accepted presumption that leakage took place. Therefore, any null result produced can be interpreted as an absence of detectable effects of leakage on the net trade position of the countries involved, which contradicts the claim about the fact that decline in emissions in rich countries results mainly from offshoring production activities.

1.5 Approach and summary of findings

The emissions data coming from Global Carbon Project 2025 [13](Friedlingstein et al., 2025) is used with the covariates from World Bank and UN for the panel of 119 countries since 1990 till 2023 years. Treated group makes up 34 Annex B countries, and comparison group includes 85 countries that do not have binding targets. Estimation of the commitment effect is done using two-way fixed-effects difference-in-differences (DiD) model, event study, joint test of pre-treatment trends, and many robustness checks. The analysis shows no significant evidence of carbon leakages induced by Kyoto Protocol and in the national gap measures. The value of commitment effect is equal to -0.070 (standard error 0.073). This means that the result is contrary to H_1 and not statistically different from zero. The values of the secondary outcome and robustness tests are insignificant either. Coefficients in the event study measured against the year before the implementation are mostly negative in post-treatment years, which goes against the hint of leakage. There is also important limitation of this thesis analysis, since the corresponding test shows the presence of pre-treatment effects.

1.6 Structure of the thesis

Chapter 2 reviews the literature on the EKC, trade and pollution havens, consumption-based accounting, carbon leakage, evaluations of the Kyoto Protocol, and recent work on difference-in-differences methods. Chapter 3 covers data collection, data integration, feature engineering, the exploratory analysis and the econometric approach. Chapter 4 presents and discusses the results. Chapter 5 concludes, and Chapter 6 gives recommendations for policy and further research.

2 Literature Review

The chapter summarizes the research works upon which the thesis is built. It starts with the Environmental Kuznets Curve and its critiques, and then proceeds to trade and pollution as well as the techniques used for measuring embodied emissions. Then it goes on to discuss the theory and evidence regarding carbon leakage, the evaluation of the Kyoto Protocol, and the new methodology of the difference-in-differences approach. The chapter ends by specifying the gap that this thesis is intended to fill.

2.1 The Environmental Kuznets Curve: origins and early evidence

The Environmental Kuznets Curve derives its name from the theory by Kuznets that states that income inequality and income per capita has an inverse direct relationship. The environmental Kuznets curve was formed in the early 90s due to the work suggesting that trade liberalization would have an influence on the environment.[15] Grossman and Krueger (1991) observed the impact of the North American Free Trade Agreement and found that urban areas with a lot of pollution and smoke were rising along with the income in many countries. The question, which has to do with different types of pollutants, was analyzed in their following work done in 1995. The authors concluded that economic growth leads to environmental degradation. Later studies began showing that this model applies to some types of pollutants because it is not universal. It provided evidence of inverted-U for some of the air pollutants though the levels of income at which the turning points occurred were too high for the countries to achieve. Another research proved the existence of inverted U for local air pollutants and that some factors such as availability of clean water improved together with the increase of income. [17] Holtz-Eakin and Selden (1995) built an EKC for carbon dioxide which was extended too much in time ahead. Pollutants which cease to be harmful after a certain delay are political in nature. The theoretical rationale for the inverted-U has been classified into three types of effects by [12] Dinda (2004): emissions increased due to the growth of output (scale effect); moving from manufacturing to services reduces emissions (composition effect) and wealthier countries have the capability to invest in cleaner technology and support stricter regulations (also known as technique effect). The role of international trade comes in through the composition effect, since what the country produces can become cleaner and not the activity which is subject to consumption.

2.2 Criticism of EKC's econometrics and new approaches

EKC literature has met numerous criticisms regarding its econometrics. In Stern's words (2004), early studies are based on faulty statistical grounds. The findings may change based on the sample, functional form, and method of estimation. The serial dependence of data is taken as given, and the omission of the important variables may lead to false conclusions on the existence of inverted-U shape. Wagner points out that standard EKC regression applicability of non-linear transformation like squared logarithm of GDP to a unit root variable, while standard statistical approaches applied to this case can lead to incorrect results. Nevertheless, recent studies take a new look at the problem by applying larger samples and employing new variables. For instance, Wang et al. (2024)[23] study 147 countries in the period from 1995 to 2018. They show that the inverted-U shape was

present in the case of emissions of high-and upper-middle-income countries but at the same time reveal that trade protectionism plays a distorting role in the linkage since this slows down the process of diffusion of clean technology. According to Wang, Li and Li (2025)[23], twelve variables related to economic, technological, institutional, social, and resource variables concerning 214 nations revealed an N-shaped curve according to which emissions decline up to a certain degree of income but start rising later. According to Li (2023)[18], the effect of the share of non-fossil fuel energy and total productivity is responsible for the angle of this curve, and it is combined with the Solow-type growth model. The issue is treated generally with the understanding that what is called EKC is only valid in certain conditions and not valid in others.

2.3 Trade, creation of pollution havens and composition of output

The concept that the negative slope of the EKC depends on the idea of relocation is closely connected with the pollution haven hypothesis, which states that environmental regulation levels among countries are essential for achieving comparative advantages. According to Copeland and Taylor (2004)[9], they distinguish between pollution haven and pollution haven idea. It suggests that environmental regulations will create high costs for the environment and change both trade flows and location of factories, while the pollution haven idea claims that increasing trade will move pollution-generating industries from one location to another in the course of time. The authors find evidence of the first hypothesis but nothing to prove the second one. Hence, while the rules regarding regulations do matter, they play only a minor role in the overall economy. Endogeneity complicates the study of the pollution effect. Earlier studies show no correlation between regulatory costs and trade. As claimed by Levinson and Taylor (2008), the highest degree of net imports from Mexico and Canada is related to US industries with the biggest increase of abatement costs once the influences of unobserved heterogeneity and endogeneity of abatement costs are considered. The influence of trade on the concentration of Sulphur dioxide can be divided into scale, technique, and composition effects according to Antweiler, Copeland, and Taylor (2001)[9], and according to them, the negative effects of free trade prevail. It explains that although there is indication of trade's contribution to the slope of EKC for some types of pollution, this influence is insignificant. In their comprehensive overview of the influence of environmental regulation on competitiveness, Dechezleprêtre and Sato (2017)[11] arrive at the conclusion that there are detectable effects of regulated trade and impacts on investment location. In this thesis, the pollution haven literature provides two conclusions. First, the relocation of production is plausible, as regulation does influence trade at the margin. Second, the impacts are insignificant in comparison with the other determinants of trade.

2.4 Consumption-based emissions accounting

Consumption-based accounting supplies the measurement the pollution haven literature was missing. Production-based, or territorial, accounting, which national reporting to the UNFCCC uses, assigns emissions to the country where they physically occur. Consumption-based accounting assigns them to the country where the resulting goods and services are finally consumed. The difference between the two equals the emissions embodied in imports minus those embodied in exports. Wiedmann (2009) reviews the

multi-regional input-output methods used to produce these estimates. Peters and Hertwich (2008)[19] show that emissions embodied in trade are large relative to national totals, and that how they are treated matters for dividing responsibility under climate agreements. Davis and Caldeira (2010)[10] estimate that about a quarter of global CO₂ emissions in 2004 were embodied in internationally traded goods, mostly flowing from China and other emerging economies to the United States, Europe and Japan. Peters et al. (2011)[19] follow these flows over time. They find that net emission transfers from developing to developed countries grew substantially between 1990 and 2008, by more than the reductions developed countries achieved under the Kyoto Protocol. This finding is often cited as evidence that apparent progress in rich countries was partly cancelled out by rising emissions elsewhere. Consumption-based accounts have weaknesses of their own. They rely on input-output tables with coarse sector detail, on assumed emission intensities for each exporting country, and on choices about how to treat re-exports and international transport. Kander et al. (2015) argue that pure consumption-based accounting penalises countries whose export sectors are clean, and they propose a technology-adjusted version. The Global Carbon Project, whose data this thesis uses, publishes yearly consumption-based estimates for about 120 countries next to its territorial series (Friedlingstein et al., 2025)[13]. Because both series come from one source, the gap between them reflects trade and is not distorted by differences in what each dataset covers.

The consumption-based accounting provides the measurement that has been missing in the pollution haven literature. The production-based accounting, which is used in national submissions to the UNFCCC, allocates emissions to the country where they are generated. The consumption-based approach is different. It allocates emissions to the country where the goods produced are consumed. The difference between two approaches is emissions embodied in imports and the emissions embodied in exports. The methods for making such estimates are reviewed by Wiedmann (2009). According to Peters and Hertwich (2008), emissions generated from trade should be evaluated in relation to the emissions produced by countries. These authors provided an analysis of the level of emissions generated in trade from different countries including China and how they are accounted for in climate change agreements. For instance, Davis and Caldeira (2010)[10] present the scheme according to which trade represents a coordinate fourth of the total level of CO₂ emissions involved in global trade. As noted by Peters et al. (2011)[19], however, the flow of carbon emissions from developing nations to developed nations significantly increased in the 1990s as compared to the 2008 emissions level produced through the Kyoto Protocol implementation. Consumption-based approaches also face certain criticism due to their reliance on crude input-output tables and assumptions about the emissions produced in terms of certain generating entities, as well as methods of counting trade re-exports and international transport. Kander et al. (2015) demonstrate that pure consumption-based studies must penalize countries with less developed export industries. Global Carbon Project offers its own addressing of the problem and proposes that the international community utilize a modified version of emissions accounting.

2.5 Carbon leakage: theory, simulation, and post-event verification

The idea of carbon leakage is related to the increase in emissions in countries that are not subject to regulations that are being implemented in countries where these regulations exist. There are three different ways that carbon leakage can occur. Firstly, regulated

companies can lose their share in the market due to competition from companies that are not regulated. Secondly, companies might start building capacities in places that do not face regulations. Lastly, the decrease in the demand for fossil fuels in countries that have regulations might lead to a drop in prices in the world and result in an increase in consumption elsewhere. The term “leakage rate” normally refers to the rise in emissions in other countries in relation to a country in which emissions reduction actually occurred. Most of the research on carbon leakage is based on the use of computer models. According to Babiker (2005)[3], depending on certain conditions (for example, some market structures) the level of leakage may exceed 100, and the implementation of climate policies in one country can result in an increase of emissions in the world. Branger and Quirion provided evidence that the findings of different carbon leakage studies vary remarkably and the vast majority of them show that the level of emissions leakage is normally between 5 and 25 percent[5]. However, it is important to understand that the mentioned figure is a prediction, and thus, it has to be verified. As for the post-event verification of data, it is not so easy to investigate. For instance, certain researchers found that the implementation of emissions trading in the European Union didn’t lead to the rise of emissions, with the evidence provided referring to the same. Another research by Dechezleprêtre and Sato (2017)[11] prompts the same conclusion where environmental regulation is concerned. Although this could be possible in reality, it has also had an influence on policy design. Sectors that are thought to be potentially affected by leakage under the EU Emissions Trading Scheme have been allocated free allowances rather than auctioning them, which limits their exposure to carbon prices but puts pressure on them to limit their emissions. Europe’s Carbon Border Adjustment Mechanism established in 2023 presents a different approach. Importers of cement, iron and steel, fertilizers, electricity, and hydrogen are obliged to report emissions that are included in price at that time, and from 2026 they will be required to pay for these emissions while free allocation to the same domestic sectors will gradually be phased out. However, both mechanisms are based on the assumption that leakage is indeed a risk for these sectors and. Ex-post literature examined above indicates that risks of leakage were lower in the past than appeared according to the models, although carbon prices have increased dramatically as compared to the period of Kyoto protocol.

2.6 The Kyoto Protocol as an object of evaluation

This study departs from Aichele and Felbermayr[1] in three ways. It looks at national net positions instead of bilateral sector flows, so it asks whether leakage was big enough to change a committed country’s overall net trade in embodied carbon, which is a different question from whether leakage happened in particular trade relationships. Its panel runs to 2023 and so covers the whole first commitment period and the years after it. And it applies the pre-trend diagnostics developed in recent difference-in-differences research, reviewed next.

It is important to note that the way the protocol was designed has an impact on its leakage effect. During the first commitment period, which was from 2008 to 2012, the countries of Annex B agreed to cut emissions by approximately 5% down to the level of emissions for 1990. Each country had its own individual targets, ranging from cuts in nearly all European countries to allowed increases for a few, such as Australia and Iceland. Several economies in transition had targets greatly in excess of actual emissions they recorded when they collapsed industrially in the early 1990s, so the commitments

were not binding in practice. The Protocol has also established flexible mechanisms. Under the Clean Development Mechanism, the countries listed in Annex B could comply with part of their commitments by paying for emission-reduction projects in Annex B countries. This works against leakage, because this reduces emissions in those countries that are uncommitted and have leakage. A second commitment period was agreed on in 2012, covering 2013-2020 where, by this time, Canada had already withdrawn, and Japan, New Zealand and Russia did not commit to any new targets. Hence, the intensity of compliance varied across countries and dropped after 2012. Because of a clear distinction between committed and uncommitted countries, the Protocol has its own body of evaluation literature. The findings regarding its influence vary. For instance, Grunewald and Martínez-Zarzoso (2016)[16] reported a negative relationship between binding commitments and CO₂ emissions for the respective countries. Almer and Winkler (2017) carried out an analysis based on synthetic control methods for several countries and did not discover any statistically significant reduction of emissions. According to Maamoun (2019), the Protocol has led to a few per cent reduction of greenhouse gases when compared with counterfactual and therefore can be called a hidden success. The study of Aichele and Felbermayr most closely resembles the above-mentioned work in content. In their initial research, Aichele and Felbermayr (2012)[1] demonstrate that commitment to Kyoto was associated with reductions in domestic emissions but did not lead to considerably lowered carbon footprints of the nations, which indicates that the increases in embodied imports canceled out the decreases in domestic emissions. Later in their second study, Aichele and Felbermayr (2015) analyzed the role of the commitments in increasing embodied carbon imports from noncommitted nations by applying bilateral sector-level data on carbon content of trade in a gravity model framework. These two pieces of research are the only ex-post evidence of the existence of leakage caused by Kyoto and this paper compares its results with the results of Aichele and Felbermayr. This research has some differences in relation to what was conducted by Aichele and Felbermayr on several levels. First of all, it is conducted at the national level, while the studies of Aichele and Felbermayr were conducted at the bilateral sector level. At the national level, the researchers analyze whether leakage could lead to changes in the countries' net trade with the embodied carbon. The second difference is that this research studies the entire period of the first commitment period, as well as years that follow it up to 2023. The third difference is that the method of analysis used in this paper utilizes pre-trend diagnostics developed by authors of recent difference-in-differences research.

2.7 Methodological developments in difference-in-differences

Difference-in-differences compares how outcomes change in a treated group with how they change in a comparison group. It rests on the parallel trends assumption: without treatment, the two groups' outcomes would have moved in the same way (Angrist and Pischke, 2009)[2]. This cannot be tested for the post-treatment period, but it can be checked indirectly by asking whether the groups moved together before treatment. Several recent studies contribute to the approach in which these estimators should be determined and understood. Bertrand, Duflo and Mullainathan (2004)[4] presented the argument that the presence of serial correlation in the outcomes in a panel leads to under-specified regression standard errors, so the authors recommended conducting clustering according to the level of treatment implementation. Cameron, Gelbach and Miller (2008) demonstrated that – in the case of few clusters – cluster-robust inference can be misleading, so

the authors gave their wildcard clustering algorithm. A large number of studies regarding staggered adoption showed that good two-way fixed-effects estimators are biased in terms of treatment timing and unit-level effects because early units become controls for late units (Goodman-Bacon, 2021[14]; de Chaisemartin and D’Haultfoeuille, 2020[7]; Sun and Abraham, 2021). Callaway and Sant’Anna (2021)[6] developed estimators which do not need to make use of the above comparison. The discussed problems do not occur in the case of treatments in the same year that is considered in this paper, and the results obtained are similar to those produced by traditional estimators. The second line of work deals with pre-trending tests, which were shown by Roth (2022)[21] who concluded that traditional tests of pre-treatment coefficients have very low power against important violations and the approach when the testing has to be passed for conducting inference leads to distorted results. Rambachan and Roth (2023)[20] have discussed sensitivity testing in their paper, where the author introduces bounds on post-treatment deviations from parallel trends and then assesses the effect of those bounds on the resulting achievements. Roth et al. (2023)[22] have further elaborated on the above pursuit. The answer for practitioners is a simple lesson: if the pre-trend test fails it should be reported, and its effect analyzed. Adjustment of the data until one ends up with a positive test result instead of admitting the failure hides the problem. The third aspect of the problem is the choice of outcome. Chen and Roth (2024)[8] state that in cases when the outcomes are either zero or very close to zero, one may use transformations of the data, such as log of one plus outcome and inverse hyperbolic sine, where estimated values depend on the units of measurement. In this respect, many emissions operations are small, and this is why the thesis mostly reports share as an outcome measure leaving the inverse hyperbolic sine as a measure of robustness with non-zero value of scaling constant.

2.8 Research gap

It should be noted that each of the above-mentioned lines of literature does not give full answers to the research questions. The papers on EKC show that emissions of developed countries have stabilized or declined, but having predominantly use production data they may not determine whether the decline is real, since there have been no changes in consumption. The literature on environmental leakages does not answer the question, but rather shows some predictions with a very limited number of ex-post studies. The authors of the ex-post researches are Aichele and Felbermayr who find the evidence of emissions leakages induced by Kyoto Protocol on the bilateral level. The gap to be filled in this thesis is the assessment of emissions leakages in countries with binding commitments in terms of the net value of embodied carbon, which is done on the basis of production and consumption statistics from one relevant source during the period of time that includes the first commitment period and afterwards.

3 Methodology

This chapter explains how the data were collected, merged and transformed, what the exploratory analysis showed, and how the econometric design follows from it. All work was done in Python with pandas, NumPy, statsmodels and linearmodels, across three notebooks: `data_integ.ipynb` for data integration, `thesis_EDA.ipynb` for feature engineering and exploratory analysis, and `thesis_estimation.ipynb` for estimation.

3.1 Data Collection

The analysis needs four kinds of data: production-based emissions, consumption-based emissions, economic and demographic covariates, and the treatment assignment. Table 1 lists the sources. The emissions data are from the national fossil carbon emissions

Table 1: Data sources

Variable	Source	Coverage	Unit
Territorial (production-based) fossil CO ₂	Global Carbon Project 2025, national emissions workbook	1850 to 2024	MtC, converted to MtCO ₂
Consumption-based fossil CO ₂	Global Carbon Project 2025, national emissions workbook	1990 to 2024	MtC, converted to MtCO ₂
GDP per capita	World Bank, World Development Indicators	1990 to 2023	PPP, constant 2021 international dollars
Trade openness	World Bank, World Development Indicators	1990 to 2023	Exports plus imports, % of GDP
Population	United Nations, World Population Prospects	1990 to 2023	Persons
Annex B status	UNFCCC, Kyoto Protocol Annex B	Not applicable	Binary

workbook published alongside Global Carbon Project 2025 (Friedlingstein et al, 2025), providing information on territorial emissions since 1850 and consumption-based emissions since 1990, both in million tonnes of carbon, for every country and regional totals. Both data series refer to emissions from fossil fuel combustion and cement production. The conversion from carbon to CO₂ was conducted by multiplying by 3.664, the ratio of molecular weight of CO₂ to atomic weight of carbon. EDGAR dataset was evaluated and then dismissed. EDGAR covers total greenhouse gas emissions in CO₂ equivalents. The problem of mixing different scopes arises here: if territorial emissions from EDGAR are deducted from consumption-based emissions from Global Carbon Project, the resulting difference captures partially what each data set includes. GDP per capita at purchasing power parity in constant 2021 international dollars, as well as trade openness values, comes from World Bank World Development Indicators. PPP figures at constant prices allow comparing incomes across countries and years, devoid of distortions made by changes in exchange rates and inflation which is impossible with current US dollar figures; total world population information is based on UN World Population Prospects. The approach is in line with a list of countries with implemented quantified emission restrictions given in the Annex B of the Kyoto Protocol. The US signed the Protocol but did not ratify it, thus it had no binding obligation. Based on Aichele and Felbermayr (2015)[1], it is regarded as not treated in terms of emissions. The study covers the period from 1990 when consumption series starts, to 2023, since the 2024 figures are preliminary.

3.2 Data Integration

In the emission workbook, countries are structured as columns and years as rows, with regional aggregates, bunker fuels and a statistical difference term alongside the respective countries. The integration process, which was completed in five steps, began with the loading of the emissions sheets, before reshaping them to the long format having one observation per country and year. Any row entries that were not actual countries were removed, thus ensuring that only aggregates such as the world aggregate, continental totals, as well as international and shipping entries were reassigned. The next prospect of the integration called for determining ISO3 codes for countries. All names were mapped to ISO3 codes through the `country_converter` package, thus providing a unique identifier for those countries. The merger that followed, took 1990 to 2023 as the period of focus and called for the presentation of values for both series in the long format, where only one ISO3 code was unique for a country thus avoiding naming convention mismatch in cases where the same country is represented through different names. Instructions on the third stage stipulated the merger of both consumption and territorial series for the years of interest, which also required reshaping both datasets from the wide format to long one for a more feasible merger. One challenge that was encountered during the merger was a necessity to make use of the same name for Tanzania in two entries. Yet, this issue was resolved as Tanzania values for both years were summed up and resulted in the single observation on Tanzania. Subsequently new variables for the advanced treatments were created, including `annex_b` for all countries that belong to Annex B and were given 1 in the relevant variable, while getting 0 for all other countries in the sample. The creation of the treatment variable `first_treat` was completed in the same way, where countries were assigned their first date of treatment in their respective columns with both groups of countries getting either 2005 or 0 for first treatment based on Callaway and Sant'Anna (2021)[6]. Later, the merger of the panel was conducted to ensure the correctness of the merged data, confirming that the values in the dataset are in line with the values found in the studies conducted. That confirmation showed that China was producing more emissions than it consumed, while the United States has been presenting its data in a positive way. The end result was a panel with 4032 rows that included data for 119 countries for the years of 1990 to 2023. However, only 34 countries in Annex B could be found in the panel alongside 85 other countries, providing evidence of a balanced panel and confirming the supremacy of developed countries in this sample. The dataset, though, faced some challenges such as the missing data for GDP along with the negative data for trade openness.

3.3 Feature Engineering

This is how the variables were constructed. The variable `P` refers to production emissions, and `C` refers to consumption emissions in million tonnes of CO₂. The variable `N` stands for the total population of the country. The net transfer is derived from the consumption emissions from production emissions based on the formula given below:

$$\text{transfer} = C - P \quad (1)$$

A positive net transfer indicates that the country imports more carbon emissions than it produces while a negative value indicates that it exports more than it imports. The most important result is the gap share, which is given by the formula below:

$$\text{gap_share} = \text{transfer}/P \quad (2)$$

The gap share represents the net transfer as a proportion of domestic production emissions. It is scale-invariant, so countries regardless of size both contribute similarly. The denominator is non-zero throughout the analysis; the minimum production value observed was 0.451 Mt. In addition to the version of the gap share described above, I also created a gap share in which consumption emissions are taken as the denominator, but it is numerically unstable as consumption emissions can obtain values close to zero and even negative. This is the case for Panama, which has recorded negative consumption emissions in 2006, 2008, and 2016, which is due to how bunker fuels and canal traffic are recorded in the input-output analysis. Consumption emissions for these observations are set to missing values which signifies that only the positive values are retained. Panama is not included in the treatment group, so the analysis is performed only on the treated group. The secondary outcome variable is the net transfer per capita expressed in tonnes of CO₂:

$$\text{transfer_pc} = \frac{(C - P) \times 10^6}{N} \quad (3)$$

Emissions are in million tonnes, so multiplying by 10⁶ gives tonnes in the numerator and the result is in tonnes per person. Production and consumption emissions per capita were built the same way. The robustness outcome is a scaled inverse hyperbolic sine of the net transfer:

$$\text{iht_scaled} = \text{arcsinh}\left(\frac{\text{transfer}}{\theta}\right) \quad (4)$$

The inverse hyperbolic sine will behave as logarithm at larger values. It is defined also when the argument is equal to zero or negative. The requirement is generally fulfilled by transfer distributions which have heavy tails and both positive and negative values. Chen and Roth provide evidence that using the inverse hyperbolic sine requires careful treatment when there are plenty of observations close to zero. In this panel, 45.3% of observations have absolute transfers below 5 Mt and 17.0% below 1 Mt and thus the authors' argument is applicable. Therefore, net transfers were divided by some scaling constant $\theta = 3.755$ Mt which is equal to the median absolute transfer before the treatment takes place. Log emissions per capita were calculated for EKC analysis but for the reasons explained in Section 3.4 they are not used in the model. Log GDP per capita values were centered and then square and cube of the centered variable were calculated to eliminate multicollinearity. Finally, let me explain the treatment variables. The form post indicates that the treatment started in 2005. The variable `treat_post` equals `annex.b` times `post`. Relative time means years before and after 2005. For that study, the relative time was binned at the minimum. The year 2004 is assigned as the reference year before the treatment.

3.4 Exploratory Data Analysis

The exploratory analysis had four objectives. The first was to identify the distribution of the variables. The second was to determine the similarities between treated and comparison groups prior to treatment. The third was to find outcomes with parallel trends before treatment, and the fourth was to identify the income-emissions relationship behind the research question.

3.4.1 Distributions

The net transfer is not a normal distribution. It has a skewness of -7.35 and a kurtosis of 82.5. The negative skew is due to the presence of a few large net exporters, mainly China, which has transfers that are much larger than those of most countries. However, the majority of the observations are relatively small. The combination of the heavy tail and the large distribution around zero causes the primary outcome to be a scale-free share, which is why the inverse hyperbolic sine is scaled accordingly. The consumption-denominator gap share takes on extreme values whenever consumption emissions are low, and has a minimum value of -57.2. For the purposes of exploratory checks, this variable was winsorised at its 1st and 99th percentiles. The production-denominator gap share, the main outcome, is less affected since production emissions cannot be negative.

3.4.2 Covariate balance before treatment

Table 2 displays the normalised differences between treated and comparison groups prior to 2005. Normalised difference refers to the difference between the means of the two groups divided by the square root of the average of the variances of both groups. According to Imbens and Wooldridge (2009), the absolute value of above 0.25 implies that linear adjustment may be unreliable. Table 3.2 illustrates the pre-treatment normalized differences between the treated and the comparison groups from 1990-2004 and shows that there is not a great deal of difference with respect to trade openness but rather a major gap regarding GDP and emissions per capita. Countries in Annex B show higher income levels and emissions per capita as compared to countries in the comparison group. However, there is some similarity as far as trade openness is concerned, and the normalized difference in respect of gap share is rather modest (only 0.051). Nevertheless, the country's fixed effects cancel out differences in levels, so that there are no problems with estimation caused by the imbalance in the levels. Now the question is whether or not the outcomes of the groups have changed in similar ways, but this gap in income level is quite serious and indicates that the comparison group is structurally different from the groups treated.

Table 2: Pre-treatment normalised differences, treated versus comparison group (1990 to 2004)

Variable	Normalised difference
GDP per capita	1.038
Consumption emissions per capita	1.022
Production emissions per capita	0.702
Trade openness	0.100
Gap share	0.051

Note: The normalised difference is the difference in group means divided by the square root of the average of the two group variances, computed on pre-treatment observations (1990–2004). Values above roughly 0.25 in absolute terms suggest that linear adjustment may be unreliable.

3.4.3 Pre-treatment trends

For the purpose of evaluating each of the potential outcomes, I performed one of the linear pre-trend tests with respect to the chosen outcomes from 1990 to 2004 and used country fixed effects and clustered standard errors. Table 3 provides the results.

Table 3: Linear pre-treatment trend tests (1990 to 2004)

Outcome	Differential trend per year	<i>p</i> -value	Assessment
Gap share (consumption denominator, winsorised)	+0.0047 (SE 0.0043)	0.279	No significant differential trend
Net transfer per capita	Not reported	0.12	No significant differential trend
Log consumption emissions per capita	−0.0173	0.003	Differential trend

Note: Each row reports the coefficient on a treatment-status interaction with a linear time trend, estimated on 1990–2004 data with country fixed effects and standard errors clustered by country.

Thus, the results confirm the presence of a clear differential trend. Prior to the implementation of the Kyoto Protocol, consumption-based emissions exhibited a decline of approximately 1.7 percent annually when compared to a control group. Since the result in emissions would indicate that the pre-existing divergence was actually attributed to the treatment, those results were omitted as effects. The gap measures did not reveal any significant linear trend and as a result, were selected as outcomes. There are two stipulations. The exploratory study employed an alternative gap share denominator, while the primary outcome utilizing the main gap denominator. Furthermore, the linear test is designed to reveal persistent divergence at its best, although it fails to account for fluctuations in levels. The formal test used in Chapter 4 is designed to address both of those shortcomings since it uses the primary outcome and evaluates all coefficients prior to the treatment simultaneously. Chapter 4 shows that the primary outcome is also incapable of passing a more robust test.

3.4.4 The income-emissions relationship

I evaluated the EKC relationship based on both accounting methods. The results are summarized in Table 4, in which I looked at fixed-effect regression of log emissions per capita against cubic centered log GDP per capita with a combination of both country and year fixed effects. The results imply that while the linear term is positive in both equations, the higher-degree terms are negative. This creates a situation that can be described as an inverted U whose slope continues to increase in the high income regions rather than reverse. The estimated turning point for the production-based version is \$38,700, while that for consumption is higher than that at \$41,900. Turning points fall within the income range being examined by the study, which allows for incomes of about \$149,000.

The specification chosen will however affect the position of the turning point. For instance, a pooled quadratic specification with no fixed effects would result in the turning point being located outside the observed range informing us that the confidence interval is too broad to be useful.

Table 4: Fixed-effects EKC estimates (cubic in centred log GDP per capita)

	Production basis	Consumption basis
Linear term	0.660*** (0.120)	0.609*** (0.104)
Squared term	-0.268*** (0.031)	-0.230*** (0.033)
Cubic term	-0.042*** (0.013)	-0.032** (0.014)
Within R^2	0.493	0.447
Observations	3,998	3,995
Turning point (GDP per capita)	\$38,667	\$41,895

Note: Country and year fixed effects included. Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$. Turning points are the income levels at which the first derivative of the fitted cubic is zero and the second derivative is negative.

In the context of the research question, it turns out that comparing the two bases gives us more valuable information than the actual turning point on each curve. For instance, if relocation were to be the sole reason for drops in emission from sectoral production in high-income countries, consumption emissions would continue increasing even after production emissions begin to fall. In this case, the point turning point obtained from the consumption emissions curve would be equal to or greater than the turning point derived from the production curve meaning that two points are very close to each other, contrary to the presumption that emissions are driven by relocation. The findings also offer descriptive findings without showing how policies may have affected this outcome or separating the reasons behind this outcome leading to adoption of a difference-in-differences approach.

3.4.5 Further observations

Three more findings shaped the design. The gap share has a positive correlation with trade openness (+0.14), whereas the correlation is more or less zero (-0.02) at the level of the countries. This means that the correlation in the aggregated data is more of an inter-country nature than an intra-country one and, thus, trade openness can hardly be of any use as control in a fixed-effects model. Once weights are introduced in the analysis, that is, weights based on population figures only, the difference in gap shares across countries disappears, since China and India become the dominant features of the weighted analysis. It is known that in the unweighted analysis every country has an equal weight and as such sees the response of the average country in the analysis. As a final note, it is worth mentioning that out of 34 countries that have undergone treatment, there are 13 economies that used to have centrally-planned economy: they are Bulgaria, Croatia, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Russia, Slovakia, Slovenia, and Ukraine and their emissions fell considerably in early 1990s due to reasons that do not have anything to do with climate policy and that pose a considerable threat to the reliability of the treated group. Therefore, their exclusion from the final analysis needs to verification.

3.5 Econometric Approach

3.5.1 Main specification/ Base Model

Kyoto impact is evaluated using the two-way fixed-effects difference-in-differences model:

$$Y_{it} = \alpha_i + \lambda_t + \beta(\text{AnnexB}_i \times \text{Post}_t) + \varepsilon_{it} \quad (5)$$

where Y_{it} shows the variables for country i at year t , α_i shows variables for country i , λ_t shows variables for year t , AnnexB_i shows the status of each country in terms of Annex B and t refers to the time starting from 2005. The coefficient β signifies the estimate of the difference in difference of how the commitment works for the treated countries over the period following treatment, and the fixed effects associated with countries capture all the differences in time-constant variables between the countries such as disparities in income and measures of emissions. Year fixed effects capture the shock that impacts all countries regardless of their readiness to implement the Protocols such as the financial crisis or the coronavirus pandemic. In the second model, the number of GDP per capita in log will be included. The year 2005 is referred to as the year when signing the Protocol took place for all committed countries uniformly. This means that the current research uses the difference in difference design. According to Goodman and Bacon (2021), the bias of fixed effects estimation in the situation of staggered adoption has to do with the variable form of treatment, which is not the case in this regard because the fixed effects model is subject to consistent values. The uniform approach is a matter of choice and bears a certain cost. The entry into force is understood as the moment in time when obligations became binding for every party that signed the Protocol making it as a result a homogeneous event. Although individual signing dates allow somewhat more accurate treatment of such countries as Australia and Croatia, it would complicate the analysis because of the staggered timelines, and it is likely that countries that signed the Protocol later were statistically significantly different compared to those who ratified it. All regressions in this section of research are unweighted, meaning that each option is equally regarded whatever is its size, because the goal here is to see how the standard country reacted.

3.5.2 Event Study

The event study approach allows estimating the way the difference between the groups changes over time. To do this, it is necessary to apply the following event study model:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k (\text{AnnexB}_i \times \mathbf{1}[t - 2005 = k]) + \varepsilon_{it} \quad (6)$$

In this formula, k refers to the relative time, and the year before the treatment ($k = -1$, or year 2004) is not included into the model. Therefore, $\delta(k)$ indicates the treatment-comparison gap in the corresponding year compared to the gap in the year 2004. The periods until $k = -11$ are grouped into one group, and as a result ten pre-treatment coefficients emerge. If the parallel trend did exist, the prior coefficients would have to be statistically equal to zero. Hence, the corresponding test is performed.

3.5.3 Inference

The standard errors are clustered according to countries so that the correlation within the errors can be taken into account. There are 119 clusters in total, 34 of them treated.

This amount is enough for appropriate inference.

3.5.4 Robustness Checks

Four different types of robustness checks have been performed. The first point removes 13 economies in transition, leaving 21 countries for consideration. The second establishes treatment back to 2008, the first year of the commitment period instead of its start date. The third uses pseudo-treatment from 1997, the year that the Protocol was signed, using only the data predating 2005. The targets were established in 1997, so it is more correct to see them as testing anticipation instead of a simple placebo effect. The fourth eliminates each of the large countries step by step to find out if any particular country skews the results.

3.5.5 Interpreting a null result

The thesis focuses on finding an effect that may not be present, so the interpretation of insignificant results is important. If the confidence interval is narrow and centered on zero it suggests that the effect would be weak. If the confidence interval is wide, it can indicate both no and a considerable effect. The results are therefore correctly interpreted using confidence interval reports and taking into account the smallest detectable effect. With 80 per cent power and a 5 per cent significance level, the smallest true effect the design can reliably detect is about 2.8 times the standard error.

3.5.6 Threats to identification

Several disadvantages of the case make support for identification questionable, and I describe these disadvantages here to enable the reader to interpret results taking into consideration. The European Union Emission Trading System started in 2005, at the same time as the Kyoto Protocol entered into force, and since most treated countries are EU members it is impossible to separate the two cases. The targets were defined in 1997, thus firms and governments could have made their response well before 2005. Treatment status has also changed over time. Canada withdrew from the Protocol in 2012, Australia and Croatia signed it only in 2007, thus classifying all Annex B countries as treated from 2005 to 2023 is incorrect. If there is a leakage then it concerns both treated and comparatives, thus undermining the assumption of unaffected comparatives. And the comparatives differ significantly from the treated group thus making equality of trends difficult. Chapter 4 provides evidence on these challenges, and Chapter 5 describes their implications. An early version of the design applied Granger causality tests and panel vector autoregressions of rich countries and poor countries emissions. These two methods were omitted since time order does not help identify leakage. Emissions of developing countries may become higher only because of the industrialization and deindustrialization process taking place in the other group of countries. Difference-in-differences design employs a policy that changed the incentives for one of the groups of countries and kept the incentives of the other group unchanged.

4 Results

This chapter provides the description of the sample used in estimating, followed by the comparison of means of groups of observations, and then the report of the principal

findings of difference-in-differences estimate method. In the next section, the chapter mentions the event study, the formal analysis intended to test the hypothesis of parallel trends, and tests of robustness, before finishing with the comparison of the findings with the literature.

4.1 The estimation sample

The number of observations in the estimation sample is 4,029 for 119 countries. The only difference with the complete panel consisting of 4,032 observations is that the three observations of Panama with negative consumption emissions are excluded as it has not been treated. Thus, the treated group remains the same: 34 countries instead of 85 in the comparison group. In the case when log GDP per capita indicator is considered, the number of observations equals 3,995 as it is missing for Venezuela.

4.2 Descriptive comparison

Group means that were calculated before and after treatment show the first results. The main measure indicates an increase of the average gap share of the Annex B countries from 0.170 prior to 2005 to 0.305 afterwards, which signifies an increase of 0.135. As for the comparison group, the number rose from 0.149 to 0.359, which suggests an increase of 0.210. Both observations show that groups were more active in importing emissions than they were in producing carbon, however, the reaction of the comparison group was larger than that of the treated one, which made the final result for the first difference-in-differences negative.

Table 5: Naive difference-in-differences in group means

Outcome	Naive DiD estimate
Gap share (primary)	−0.0750
Net transfer per capita, t/person (secondary)	+0.0432
Scaled IHS of net transfer (robustness)	−0.0044

Note: Each estimate is the change in the treated group’s mean from the pre-treatment to the post-treatment period, minus the corresponding change in the comparison group. No fixed effects or controls are included.

This method of difference-in-differences reveals the main conclusion already in the naive estimates because disclosers indicate the opposite tendency than predicted by leakage. The secondary output is more positive, however, much smaller than the primary one and the output related to the robustness test is near to zero in principle. The committed outcome order predicted that the primary, together with secondary outcomes, could differ in direction, with the gap share being designated as a primary outcome.

4.3 Main findings

Table 6 presents the findings based on the two-way fixed-effect models estimating the Kyoto Protocol’s commitment effect on all outcomes.

None of the three estimates appears to be statistically significant. For the primary outcome, the point estimate of −0.070 is approximately 41% of the treated group mean

Table 6: Two-way fixed-effects difference-in-differences estimates

Outcome	β	SE	p-value	95% CI	Pre-mean (treated)
Gap share (primary)	-0.0702	0.0732	0.338	[-0.214, 0.073]	0.170
Net transfer per capita (secondary)	+0.0527	0.3478	0.880	[-0.629, 0.734]	1.163
Scaled IHS (robustness)	+0.0019	0.1924	0.992	[-0.375, 0.379]	1.002

Note: Country and year fixed effects included. Standard errors clustered by country (119 clusters). $N = 4,029$. Treatment is $\text{AnnexB}_i \times \mathbf{1}[\text{year} \geq 2005]$. The net transfer per capita is measured in tonnes of CO₂ per person; the gap share and the scaled inverse hyperbolic sine are unitless. Pre-means are treated-group means over 1990–2004.

gap share, and is opposite to the leakage statement. The 95% CI extends from -0.214 to 0.073 . Secondary and robustness estimates are almost zero regarding the respective standard errors. The CI is still relatively wide. With the standard error being equal to 0.073 , the minimum detectable impact, assuming the 80% power and 5% significance is equal to around 0.20 , which is already larger than the pre-treatment mean gap share of the treated group. The design is able to eliminate the option of a very high leakage, such as doubling the gap share. Although the moderate effects are still possible in either direction, it can be summarized that the assumption of no effect is valid and economically meaningful outcomes cannot be classified as impossible.

Table 7: Estimates controlling for log GDP per capita

Outcome	β	p-value
Gap share (primary)	-0.0739	0.281
Net transfer per capita (secondary)	+0.0221	Not Reported
Scaled IHS (robustness)	-0.0188	Not Reported

Note: Country and year fixed effects included. Standard errors clustered by country. $N = 3,995$ (Venezuela excluded because GDP per capita is unavailable). Income may itself respond to climate policy, so this specification is reported as a sensitivity check rather than as the preferred estimate.

When controlling for log per capita GDP in the model, the estimates remain almost the same (See Table 7). While interpreting the results it should be kept in mind that there are two important issues associated with the controlled specification. One is that it is based on a smaller sample size, so it is invalid for the comparison with the previous finding (source) in terms of being based on the same data set. It is worth mentioning that income has also been measured as a post-treatment variable responding to climate policy which creates a problem of using it as a reliable control variable. For both factors this specification is a sensitivity analysis. It is not the optimal estimate.

4.4 Event Study

The event study coefficients for the two main outcomes shown in figures 1 and 2. For the gap share outcome, every coefficient in a post-treatment period is negative, and the average is -0.079 , compared to the estimate of the two-way FE estimator of -0.070 . The coefficient eight years after the treatment, -0.160 (standard error = 0.075), is significant

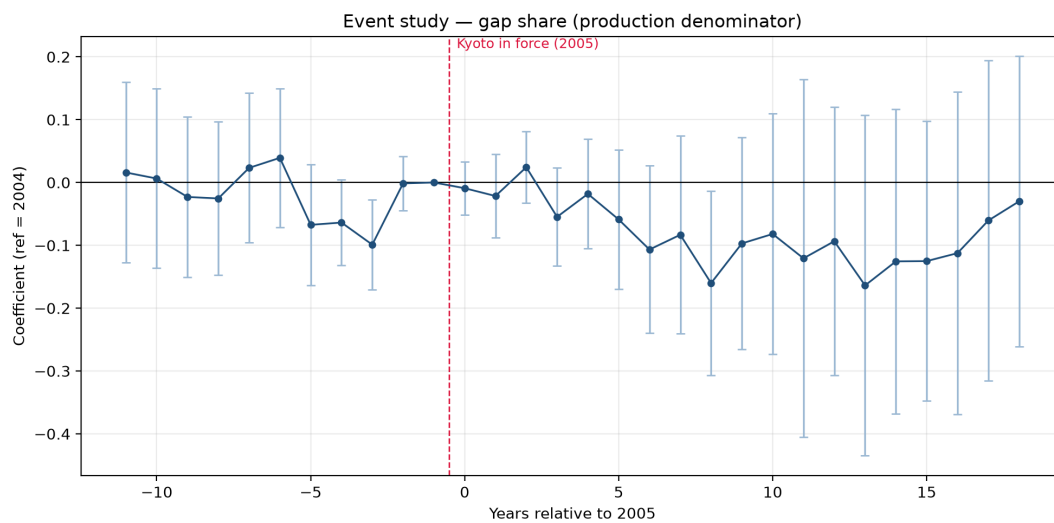


Figure 1: Event-study coefficients for the gap share, with 95% confidence intervals (reference year 2004).

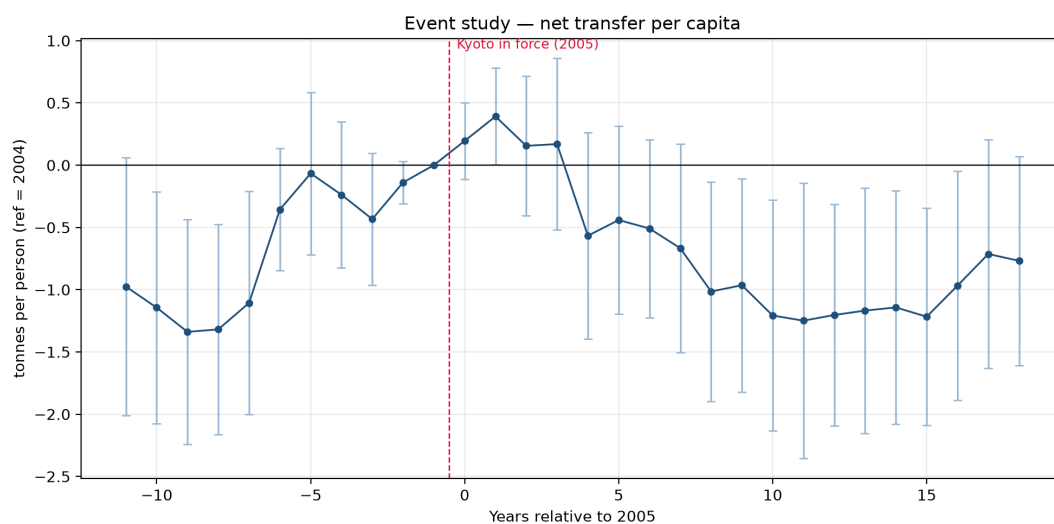


Figure 2: Event-study coefficients for net transfer per capita, with 95% confidence intervals (reference year 2004).

by itself. Before the treatment, most coefficients are of low value, except for the coefficient three years before the treatment, which was -0.099 and significant in isolation. The net transfer per capita is behaving in a different way and needs caution. For the years ten to seven before the treatment ($t=1995-1998$), the pre-treatment values are high and negative, between about -1.1 and -1.3 tons per person and statistically significant. After the eighth year following the treatment, the average post-treatment values are also negative at the level of about -1.0 to -1.2 tons per person, with the standard error being about 0.45 , with several significant isolated values. The average net value is -0.678 . This seems to contradict the two-way fixed-effects estimate of $+0.053$. The reason for it is a reference period difference. The two-way fixed-effects estimator compares post-treatment years with more years and pre-treatment average values, whereas the event study compares every single year with 2004. The treated-comparison difference was unusually low during the mid-1990s, which lowers the pre-treatment average; consequently, the post-treatment years look similar to it. In comparison to the situation right before the event took place, countries classified under Annex B have turned out to be having somewhat diminished positions as net importers per person in the later periods; this contradicts the idea of leakage. In addition, this finding underlines how much dependency the same outcome has on a specific baseline chosen in the very beginning.

4.5 Parallel Trends Assumption

The existence of the null hypothesis of the absence of all ten coefficients before the treatment is confirmed by joint F-tests.

Table 8: Joint tests of pre-treatment event-study coefficients

Outcome	F-statistic	p-value	Result
Gap share (primary)	2.20	0.022	Rejected
Net transfer per capita (secondary)	2.06	0.033	Rejected
Scaled IHS (robustness)	1.20	0.301	Not rejected

Note: Each test is of the null hypothesis that all ten pre-treatment event-study coefficients equal zero, from the specification in Equation 6. Country and year fixed effects included; standard errors clustered by country. “Rejected” denotes rejection of the null at the 5% level.

The existence of parallel pre-treatment trends can be denied at 5 per cent level, considering both main outcomes. As to the gap share, the null hypothesis was rejected on the basis of a coefficient for three years prior to treatment. The net amount of transfers per capita follows the same route and is being confirmed by the strong minus values in the mid-1990s. The only outcome which is accepted is the one which has undergone a robustness test, and this fact contradicts the results obtained after conducting an exploratory linear trend test as seen in Section 3.4.3, which has also shown no significant differences in terms of trends regarding gap share for different groups respectively. The linear test focuses on the question of whether there was any gradual divergence or not with respect to time and was applied in the context of the version linked to the denominator. Meanwhile, the joint test is organized in a way to highlight whether the difference existed at all in its movement over the time or in other words checked the outcomes of the main observation. The coefficients on average after the treatment do remain the same within four decimal points at -0.0788 and -0.6783 . Reducing the window removes the evidence

of pre-treatment divergence with the estimate left as it was. Presenting it as a way out would cover over the issue in question, thus it is decided to keep the full window and state rejection as a limitation of the estimation. Roth (2022) makes a similar argument: the fact that passing the pre-trend test after the sample change gives false security about the result. The answer should be performing the sensitivity analysis as in Rambachan and Roth (2023), asking how big a post-treatment deviation from parallel trends has to be compared to the observed deviations prior to treatment to change the conclusions. Given that the main estimate is already statistically indistinguishable from zero, the analysis will increase the confidence interval but will not change the results qualitatively. It would show the weight of the null result, however.

4.6 Robustness Checks

Table 9 presents the results on validation methods.

Table 9: Robustness checks

Specification	Gap share	Transfer per capita	Scaled IHS
Main estimate	−0.0702 (0.338)	+0.0527	+0.0019
Excluding 13 transition economies	−0.0553 (0.542)	+0.0164	−0.2111
Treatment dated to 2008	−0.0857 (0.289)	−0.2877	−0.1694
Pseudo-treatment in 1997 (pre-2005 data)	−0.0349 (0.537)	+0.5932 (0.161)	+0.2756

Note: Two-way fixed effects with country and year effects; standard errors clustered by country. p -values in parentheses. None of the estimates is significant at the 5% level. The transition-economy row leaves 21 treated countries. The 1997 row uses pre-2005 data only and tests for anticipation rather than serving as a pure placebo. The leave-one-out row reports the range of the gap-share estimate across specifications that drop each listed country in turn.

The results show that without the economies in transition there are still 21 countries treated and the main estimate is -0.055 , which is still a negative and not significant number. However the fact that the post-Soviet collapse takes place does not result in the null result, and the treatment group being smaller means loss in precision. If the year 2008 was considered to date treatment, all three outcomes resulted in negative estimates meaning that the primary and secondary outcomes do not differ in signs after all. Holding out either of the major countries from the estimation makes the main estimate stay in the range of -0.066 to -0.074 showing once again that there are no single countries responsible for the effect. The pseudo-treatment from 1997 must be examined. With respect to the primary outcome, it generates a small, statistically insignificant estimate, which is what one would expect in the absence of anticipation. On the other hand, it produces an estimate of $+0.593$ tonnes of net transfer per capita, which is not statistically significant but has an absolute value larger than any estimate produced in the case of the actual treatment. Furthermore, together with pre-trend rejection, these numbers suggest

that, for reasons not connected to the implementation of the Kyoto Protocol, treated-comparison differential for the net transfer per capita has varied a lot and therefore confirm the choice made earlier to treat the gap share as primary.

4.7 Discussion of findings

The results are consistent all over the analysis, where we have negative, respectively insignificant estimate for the primary outcome in all specifications. The results of the two-way fixed effects yielded values close to zero for the secondary outcome, while the event study made relative to 2004 demonstrates the decline. Neither of the results confirms the claim that the commitment under the Kyoto Protocol resulted in Annex B countries becoming greater net importers of embodied carbon. Aichele and Felbermayr (2012, 2015) found evidence of leakage due to Kyoto Protocol, thus showing that the results of this research contradict their findings. Nevertheless, the two studies do not contradict each other. In their work, the researchers aimed at analyzing bilateral inter-industry flows, although, in spite of concentrating leakages into certain industries or trading relations, they might not be large enough to cause any real changes in the whole country's trade position. Their samples finish earlier than those of this research, as they cover only the first commitment period and the following years, when hardly any country of the studied ones was engaged in the second period of the research. The descriptive comparison indicates that both groups in the study have also turned into larger net importers in relation to their production over the period. If the gap share of the comparison group was increasing for reasons unrelated to Kyoto, i.e., due to the advancement of global value chains or changes in trade practices, then the year effects capture the common trend, leaving the differential change to be derived from the commitment. The result is consistent with ex-post findings about the EU ETS (Naegele and Zaklan, 2019) and the analysis by Dechezleprêtre and Sato (2017), which find that climate and environmental regulation has only small trade effects. It is also compatible with the exploratory result presented in Section 3.4.4 that production and consumption emissions turn at the same income levels. If leakage were to be the main reason for the decrease in emissions in rich countries, then consumption emissions would have continued to grow after the emissions from production decreased. There are three limitations of the results. That the test for parallel pre-treatment trends has led to rejection means that the control and treated groups were moving in a different direction before 2005. The fact that the estimated sizes were based on the assumption which is contradicted by the data is a problem. The wide confidence intervals allow for moderate effects. Finally, the uniform coding of treatment ignores the fact that Australia's and Croatia's ratifications, Canada's withdrawal, and limited participation in the second commitment period in fact would have offset the estimates by causing them to be zero. Thus, one is better off interpreting the findings as a lack of evidence for leakage at the country level rather than as evidence of no leakage occurring at all. Nevertheless, in light of the fact that the main result is negative in every specification and that net transfers have decreased in comparison to 2004 in the event analysis, one cannot conclude that Kyoto triggered any significant leakage that can be observed in national accounts.

4.8 Summary of results

All in all, the findings boil down to five points. First, in respect of the primary outcome, the change from commitment is estimated to be -0.070, with 95% confidence interval ranging from -0.214 to 0.073. This means that the sign of the effect does not agree with the theory of leakage and that the effect in itself is not statistically significant. The secondary results demonstrate figures around zero for the main specification, with the event study on the secondary outcome indicating a decline relative to 2004 for the later years after the treatment. Parallel trends are rejected with the help of joint test of pre-treatment coefficients for both main outcomes, while reducing the length of the pre-treatment period explains the rejection without changing the estimate and thus is not used. The negative effect with respect to the primary outcome is confirmed when transition and large countries are excluded, when transition in treatment dates back to 2008, and when countries are excluded from the analysis one at a time. In addition, the pseudo-treatment from 1997 indicates no advance in context of the primary outcome but a large movement in secondary outcome which is insignificant, hence the preference for the analysis of the gap share. Overall, the findings suggest that there is no indication that Kyoto agreement led to above-average imports of carbon from the countries involved in it, yet the findings are vague and the assumption tough to confirm.

5 Conclusion

5.1 Response to the research question

This study raised the question of whether binding commitments made under the Kyoto Protocol have led to Annex B countries becoming larger net importers of embodied carbon than countries without binding commitments. The study involved the collection of a panel dataset for 119 countries for the period 1990 to 2023, and no evidence for such an effect was found. The effect on the gap share, the main parameter of the study, amounts to -0.070, with a standard error equal to 0.073. Its sign is opposite to that predicted by the leakage hypothesis and shows no statistical significance. No statistically significant effect was found in relation to the secondary and robustness measures either. The primary measure remains negative when countries in transition are removed from the study, when treatment is defined in 2008, and when large economies are excluded from the sample one at a time. Despite the fact that the results are not consistent, with confidence intervals large enough to allow moderate effect, leading to rejection of the joint test of there being parallel trends for both main parameters, the conclusion is that any leakage caused by the Protocol was too minor to affect the overall trade position of compliant countries, and the results give weak support to hypothesis regarding leakage.

5.1.1 Implications Regarding the EKC Debate

This work began with a question that the EKC literature alone cannot answer: do the reduction in emissions from high-income countries reflect real decarbonization or mere relocation? The results respond to this question in two ways. The first finding of the exploratory investigation is that for all countries, the relationship of pollution emissions with income is of inverted U-shaped type and the turning points lie close to 38,700 and 41,900 dollars respectively. In case relocation was the chief cause of the decrease

in production emissions, consumption emissions should have continued growing. The second finding of the difference-in-differences analysis is that there is no proof of the influence caused by the major international climate commitment of the discussed period, which led to the increase of carbon imports by the countries with committed obligations. Neither of the findings prove that decarbonization of rich countries is completely genuine, since the evidence of the EKC is only descriptive and depends on its specification and at the same time the evidence of the difference-in-differences process is not very reliable. Nevertheless, both findings do not support the extreme interpretation of the EKC as being purely a byproduct of relocation. A more reasonable interpretation of the results is that the transfer of emissions through trade exists and is meaningful but happened more due to different reasons than climate policy, for example, due to global supply chains or unique climate commitments.

5.2 Contributions

The contributions of the paper are made in three ways. The article examines the Kyoto dilemma in detail and across a larger panel than previous research, using consumption and production data from a uniform source so that differences detected are due to trade rather than differences in coverage. It ensures an outcome order is pre-determined, presents all outcomes along with pre-trend tests, if successful or not, and does not hide the failure of a pre-trend test. It also promotes a view that despite any leakage that occurred at bilateral or sectoral level, it could not have caused any changes at the aggregate level.

5.3 Limitations

There is still a good number of limitations which affect the conclusions. Among them, the biggest issue is the rejection of parallel pre-treatment trends for both main dependent variables that means the treatment and comparison groups were on different trends prior to Kyoto Protocol implementation. The estimates depend on an assumption which is contradicted by data to some extent and no sensitivity analysis of the results has been produced. The treatment is applied in a uniform way since 2005 for all Annex B countries up to 2023 which does not account for both Australia and Croatia ratification in 2007 as well as for Canada withdrawal in 2012. It also does not differentiate the effect of Kyoto protocol and EU Emissions Trading System being launched in that same year and covering most treated countries. There are also big differences between comparison and treatment groups in terms of income level which complicates making parallel assumptions. If any leakage took place, it would impact both the treatment and comparison groups which contradicts the idea that treatment group is not affected by intervention. The data sets also bring about some limitations. Consumption based emissions are calculated based on input-output tables and there may more errors in calculating such emissions for smaller economy which has the same weight with bigger economies in the given approach. Also countries that do not submit data to the World Bank are not considered, with Taiwan being a good example. Therefore one can come to the conclusion that a nationally-based approach cannot measure leakage that happens at the sectoral level or is compensated by changes in trade patterns. Finally, the research design is limited in terms of power. Given that the minimum detectable effect on the variable of interest is around 0.20, the design allows to identify very high levels of departure from the average, but misses any moderate impact.

5.4 Final Remarks

The main lesson of the study is that it is as much about methodology as about results. A declining trend in emissions is a good stimulus for a causal story, which either claims that economic growth eliminates pollution or that pollution moves abroad. The shape of the curve cannot prove either one. To check one of these hypotheses one should have circumstances under which at least some countries would have a different response to the same situation. The findings of this research show that the Kyoto Protocol did not cause carbon leakage in national accounts, although one must admit the very limited nature of evidence suggesting that.

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